Title

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Abstract

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1 Introduction

1.1 Preliminaries

This subsection collects the notions on graphs and non-negative matrices used for both models, following the course slides.

1.1.1 Graphs

A *weighted digraph* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ has node set $V = \{1, 2, \ldots, n\}$, edge set $E \subseteq V \times V$ and a weight $a_{ij} > 0$ for each edge $(i, j) \in E$; an edge (i, i) is a *self-loop*. Throughout the report the nodes are agents, and an edge (i, j) means that agent i is influenced by agent j. This is the convention of the course slides; the tutorial by Proskurnikov and Tempo draws the edges in the opposite direction.

Definition 1.1 (Weighted adjacency matrix)**.** Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ of order n, its *weighted adjacency matrix* is the $n \times n$ matrix A whose (i, j) entry is equal to the edge weight a_{ij} if $(i, j) \in E$, and is equal to zero otherwise.

Definition 1.2 (Strong connectivity and globally reachable nodes)**.** A digraph $G = (V, E)$ is *strongly connected* if there is a directed walk from any node to any other node. A node of G is *globally reachable* if it can be reached from any other node through a directed walk.

With our convention, a directed walk from i to j means that agent i is influenced by agent j, either directly or through intermediaries. Hence G is strongly connected when every agent influences every other agent, at least indirectly, and a globally reachable node is an agent that influences the whole group.

Definition 1.3 (Periodic digraph)**.** A strongly connected digraph is *periodic* if there exists an integer $k > 1$ (the period) that divides the length of every simple cycle of the digraph; otherwise it is *aperiodic*.

Equivalently, a strongly connected digraph G is periodic if and only if the greatest common divisor of the lengths of its simple cycles is larger than 1. In particular, a strongly connected digraph with a self-loop is aperiodic, since the self-loop is a cycle of length 1.

Definition 1.4 (Strongly connected components and condensation digraph)**.** Given a weighted digraph G, a subgraph S is a *strongly connected component* of G if S is strongly connected and any other subgraph of G strictly containing S is not strongly connected. The *condensation digraph* $C(G)$ is obtained by collapsing each strongly connected component of G into a single node: for any two distinct strongly connected components S_k and S_h, there is an edge (S_k, S_h) in $C(G)$ if and only if there exists an edge $(i, j) \in E$ with $i \in S_k$ and $j \in S_h$ (self-loops are not allowed in $C(G)$). A node of $C(G)$ with no outgoing edges is called a *sink*.

A sink of $C(G)$ is a group of agents that are influenced only by members of the group itself: no agent outside the group can change their opinions.

If G contains a globally reachable node, then $C(G)$ has a unique sink.

1.1.2 Matrices

Definition 1.5 (Row-stochastic matrix)**.** A matrix $A \in \mathbb{R}^{n\times n}$ is *row-stochastic* if it is non-negative and $A\mathbf{1}_n = \mathbf{1}_n$, i.e. each row sums to 1.

Theorem 1.6 (Spectral properties of row-stochastic matrices)**.** *If $A \in \mathbb{R}^{n\times n}$ is row-stochastic, then $1 \in \operatorname{spec}(A)$ and $\rho(A) = 1$; hence $\operatorname{spec}(A)$ is contained in the closed unit disk.*

Definition 1.7 (Semi-convergent matrix)**.** A matrix $A \in \mathbb{R}^{n\times n}$ is *semi-convergent* if $\lim_{k\to+\infty} A^k$ exists and is finite.

The properties of a non-negative matrix that determine its spectrum are characterized by the graph notions above.

Definition 1.8 (Irreducible and primitive matrices)**.** A non-negative matrix $A \in \mathbb{R}^{n\times n}$, $n \geq 2$, is *irreducible* if $\sum_{k=0}^{n-1} A^k$ is positive, and *primitive* if there exists $k \in \mathbb{N}$ such that A^k is positive.

Theorem 1.9 (Irreducible matrices and digraphs)**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is irreducible if and only if G is strongly connected.*

Theorem 1.10 (Primitive matrices and digraphs)**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is primitive if and only if G is strongly connected and aperiodic. In particular, if G is strongly connected and has at least one self-loop, then A is primitive.*

2 FDG

2.1 Introduction to the FDG model

2.1.1 The FDG model

Consider a group of individuals $V = \{1, 2, \ldots, n\}$, where $x_i \in \mathbb{R}$ is the opinion that individual i holds on a given topic. The influences among the individuals are described by a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$: following the convention of Section 1.1, an edge $(i, j) \in E$, with weight $a_{ij} > 0$, means that individual i is influenced by individual j.

The French–Harary–DeGroot (FDG) model describes how each individual, or agent, updates its opinion: the new opinion is a weighted average of the opinions of the agents it is influenced by,

$$x_i(k+1) = \sum_{j=1}^{n} a_{ij}\, x_j(k), \qquad \sum_{j=1}^{n} a_{ij} = 1, \tag{1}$$

or, in vector form, $x(k+1) = Ax(k)$, where A is the weighted adjacency matrix of G (Definition 1.1). The constraint on the weights means that A is row-stochastic (Definition 1.5).

Since $x(k) = A^k x(0)$, the asymptotic behaviour of the model is determined by the powers of A. Three outcomes are possible:

- *no convergence*: $\lim_{k\to+\infty} x(k)$ does not exist for some $x(0)$; this happens when A is not semi-convergent (Definition 1.7), e.g. when A has an eigenvalue $\mu \neq 1$ with $|\mu| = 1$;
- *consensus*: $\lim_{k\to+\infty} x(k) = \alpha \mathbf{1}_n$ for some $\alpha \in \mathbb{R}$;
- *disagreement*: $\lim_{k\to+\infty} x(k) = v$ with $v \neq \alpha \mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

2.1.2 Consensus and social power

The following theorem characterizes when the FDG model reaches a consensus. Its condition (A3) involves the subgraph of globally reachable nodes: this set of nodes is always strongly connected: a walk between two globally reachable nodes only visits nodes that are themselves globally reachable. Hence its periodicity is well defined (Definition 1.3).

Theorem 2.1 (Consensus in DT averaging systems)**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. The following statements are equivalent:*

(A1) the eigenvalue 1 *is simple and all other eigenvalues μ satisfy $|\mu| < 1$;*

(A2) *A is semi-convergent and $\lim_{k\to+\infty} A^k = \mathbf{1}_n w^\top$, where $w \in \mathbb{R}^n$ satisfies $w \geq 0$, $\mathbf{1}_n^\top w = 1$ and $w^\top A = w^\top$;*

(A3) *G contains a globally reachable node and the subgraph of globally reachable nodes is aperiodic.*

If any, and therefore all, of the previous conditions are satisfied, then

(i) *w is the left eigenvector of A associated with the eigenvalue 1, and $w_i > 0$ if and only if node i is globally reachable;*

(ii) *the solution of the FDG model satisfies $\lim_{k\to+\infty} x(k) = \left(w^\top x(0)\right) \mathbf{1}_n$.*

Corollary 2.2 (Average consensus)**.** *If A is doubly-stochastic, i.e. row-stochastic with also $\mathbf{1}_n^\top A = \mathbf{1}_n^\top$, and (A3) holds, then $w = \frac{1}{n}\mathbf{1}_n$ and $\lim_{k\to+\infty} x(k) = \left(\frac{1}{n}\sum_{i=1}^n x_i(0)\right) \mathbf{1}_n$.*

The entry w_i is called the *social power* of agent i: it is the weight of the initial opinion $x_i(0)$ in the final opinion of the group.

Point (i) has a direct social interpretation. Since the consensus value $w^\top x(0)$ is the same for every agent, the initial opinion of agent i can contribute to it only if it reaches *all* the agents. If i is not globally reachable, there is at least one agent that is influenced by i neither directly nor through intermediaries: the opinion of i never reaches that agent, so it cannot enter the common value, and $w_i = 0$. Conversely, the opinion of a globally reachable agent reaches the whole group and has a positive weight. Hence the consensus value depends only on the initial opinions of the globally reachable agents.

2.1.3 Disagreement

Theorem 2.1 requires a single globally reachable group, i.e. a unique sink in the condensation digraph $C(G)$. We now consider the case in which $C(G)$ has $n_s \geq 2$ sinks. Each sink is a strongly connected component, so its periodicity is well defined (Definition 1.3).

Theorem 2.3 (Disagreement in DT averaging systems)**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. Assume that the condensation digraph $C(G)$ has $n_s \geq 2$ sinks. The following statements are equivalent:*

(A1) *the eigenvalue 1 is semi-simple with multiplicity n_s and all other eigenvalues μ satisfy $|\mu| < 1$;*

(A2) *A is semi-convergent;*

(A3) *each sink of $C(G)$ is aperiodic.*

If any, and therefore all, of the previous conditions are satisfied, then the solution of the FDG model satisfies $\lim_{k\to+\infty} x(k) = v \neq \alpha\mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

The final opinions can be described group by group. The agents of a sink are not influenced by anyone outside the sink, so each sink evolves as an FDG model on its own and, being aperiodic, reaches an internal consensus that depends only on the initial opinions of its members. Every agent outside the sinks converges instead to a convex combination of the consensus values of the sinks.

In social terms, a sink is a closed group that listens only to itself, and two closed groups never influence each other: each of them keeps its own consensus value. The agents in between listen, directly or through intermediaries, to several closed groups and settle on an intermediate opinion, closer to the groups they listen to more. As a consequence, the disagreement is bounded: every final opinion lies between the smallest and the largest consensus value of the sinks.

2.1.4 Rosetta stone

The results of this section, together with the consensus and disagreement theorems above, classify all the possible behaviours of the FDG model. The classification is summarized in Table 1, the so-called *Rosetta stone*, which translates each property of the row-stochastic matrix A into a property of the digraph G and into the corresponding asymptotic outcome; for each case, the table also reports an example digraph, its matrix and its spectrum.

The two consensus cases illustrate Theorem 2.1. In case (i) both agents are globally reachable and A is doubly-stochastic, so by Corollary 2.2 $w = (1/2, 1/2)^\top$ and the group agrees on the average $\big(x_1(0) + x_2(0)\big)/2$. In case (iii) agent 2 is influenced by agent 1 but not vice versa, so only agent 1 is globally reachable: $w = (1, 0)^\top$ and the group agrees on $x_1(0)$, whatever the initial opinion of agent 2.

Case (iv) illustrates Theorem 2.3 in its simplest form: each agent is influenced only by itself, so each agent is a sink of $C(G)$ ($n_s = 2$), both sinks are aperiodic thanks to the self-loops, and every agent keeps its initial opinion, $x(k) = x(0)$ for all k.

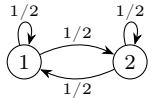
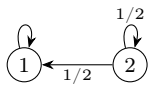
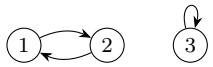
Case	Example digraph	A	spec(A)	Property of A / of G	Outcome
(i)		$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$	$\{1, 0\}$	A primitive / G strongly connected and aperiodic	consensus
(ii)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\{1, -1\}$	A irreducible, not primitive / G strongly connected and periodic	no convergence
(iii)		$\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$	$\{1, 1/2\}$	1 simple and strictly dominant, A not primitive / $C(G)$ has a unique sink, which is aperiodic	consensus
(iv)		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$\{1, 1\}$	1 semi-simple but not simple, strictly dominant / $C(G)$ has at least two sinks, all aperiodic	disagreement
(v)		$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\{1, 1, -1\}$	1 not strictly dominant / G not strongly connected, at least one sink of $C(G)$ is periodic	no convergence

Table 1: Rosetta stone of the FDG model $x(k+1) = Ax(k)$, with A row-stochastic: for each case, an example digraph (unlabelled edges have weight 1), its matrix A, the spectrum of A, the general property of A and of G, and the outcome.

2.2 Stubborness and social power in the FDG model

3 FJ

3.1 Introduction to the FJ model

3.2 Stubborness and social power in the FJ model

4 Conclusions